

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA0202

COURSE OUTCOME COVERED

CO2: Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I P Sai Bhavyatha(192424333) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Palacharla Sai Bhavyatha

RUBRICS FOR CALCULATION

Numerical Problems						
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	

Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

Annexure A

Analytical Problem Solving

Question 1: Salt-and-Pepper Noise Removal Before Edge Detection

Scenario: A grayscale image of size 1024×1024 is captured using a surveillance camera under poor lighting conditions. The image is corrupted by salt-and-pepper noise with a noise density of 0.05. This means that some pixels are randomly changed to either black or white intensity values. The image has to be preprocessed before applying an edge detection algorithm. Since edge detection is sensitive to abrupt intensity variations, the selected filter should remove noise effectively while preserving important edge details.

Answer the following:

- Identify the most suitable filtering technique for removing salt-and-pepper noise without causing significant edge blurring.
- Explain why the selected filter performs better than an averaging filter for impulse noise removal.
- If a 3×3 median filter is applied, estimate the probability that a noisy center pixel will be corrected. Clearly state your assumptions.
- Estimate the total number of noisy pixels present in the original image.
- Estimate the expected number of noisy pixels remaining after median filtering.
- Explain how the preprocessing strategy would change if the same image was affected by Gaussian noise instead of salt-and-pepper noise.

Question 2: Discretization of a Continuous Image

Scenario : An analog grayscale image has a smooth sinusoidal intensity variation along the horizontal direction. The intensity changes continuously with respect to the spatial coordinate x , while it remains constant along the y -direction. The image is defined over a spatial range of 0 to 255 in both directions. This continuous image has to be converted into a digital image for storage and further processing in a computer vision system.

Answer the following:

- Explain how the continuous image can be converted into a digital image using sampling and quantization.
- Identify the spatial variation pattern of the image in the x -direction.
- Determine the minimum sampling requirement needed to avoid aliasing in the x -direction.
- Explain what will happen if the image is sampled below the required sampling rate.
- Suggest a suitable 8-bit quantization scheme for storing the image.

- f. Discuss the effect of quantization on image quality, storage size, and intensity accuracy.

Question 3: Edge Detection in Thin Line Images

Scenario: A 512×512 grayscale image contains very thin vertical and horizontal white lines of 1-pixel thickness on a dark uniform background. Such images are commonly used in document analysis, engineering drawing interpretation, circuit diagram processing, and road-lane detection.

The objective is to detect the line boundaries accurately without losing thin edge information.

Answer the following:

- Compare Sobel, Prewitt, and Canny edge detectors for detecting 1-pixel-thick vertical and horizontal lines.
- For a horizontal edge located near pixel position (256, 256), compute the Sobel gradient response using the standard Sobel operator.
- Explain how the gradient magnitude and gradient direction help in identifying the strength and orientation of an edge.
- Discuss the role of Gaussian smoothing in the Canny edge detection process.
- Explain how Gaussian smoothing may affect edge localization, especially for thin lines.
- If the Canny threshold is reduced to half of its original value, explain the expected changes in detected edges, weak edges, and false edges.

Question 4: Feature Extraction for Handwritten Digit Recognition

Scenario: A computer vision system is being developed for recognizing handwritten digits from scanned images. The handwritten digits vary in size, orientation, stroke thickness, and writing style. Therefore, robust feature extraction is required before classification. Answer the following:

- Explain the working principles of SIFT, SURF, and ORB feature extraction methods.
- Compare SIFT, SURF, and ORB in terms of scale invariance, rotation invariance, computational cost, and real-time suitability.
- If a digit image is rotated by 45 degrees, identify which feature extraction methods can still produce reliable matching features.
- If the average number of keypoints detected per image is 120 and 50 images are processed, compute the total number of keypoints detected.
- Suggest a suitable dimensionality reduction technique to reduce the feature size while retaining 95% of the useful information.
- Explain how dimensionality reduction improves classification efficiency in handwritten digit recognition.

Question 5: Morphological Preprocessing for Character Recognition

Scenario: A binary image of size 256×256 contains scanned handwritten characters. Due to poor scanning quality, the image contains small white noise dots in the background and small black gaps inside the character strokes.

Before applying character recognition, the image must be cleaned using morphological preprocessing techniques.

Answer the following:

- Identify the morphological operation suitable for removing small white noise dots from the background.
- Identify the morphological operation suitable for filling small gaps or holes inside the character strokes.
- Explain the difference between opening and closing operations using simple imageprocessing terminology.
- If a 3×3 square structuring element is used, explain how it affects thin character strokes.
- Discuss the risk of using a large structuring element during preprocessing.
- Propose a complete preprocessing pipeline for handwritten character recognition, starting from image acquisition to feature extraction.

Question 1: Salt-and-Pepper Noise Removal Before Edge Detection

a. Most suitable filtering technique

Answer:

The **Median Filter (3×3)** is the most suitable filter because it removes salt-and-pepper (impulse) noise while preserving image edges.

b. Why median filter is better than averaging filter?

Answer:

Median Filter	Averaging Filter
Removes impulse noise effectively	Blurs edges
Preserves sharp edges	Smooths entire image
Replaces pixel with median value	Replaces pixel with average value

Therefore, the **median filter performs better for salt-and-pepper noise.**

c. Probability that a noisy center pixel is corrected

Given:

- Noise density = **0.05**
- Window = **3×3**
- Total pixels in window = **9**

Probability that surrounding pixels are not noisy:

$$\begin{aligned} P &= (1 - 0.05)^8 \\ &= 0.95^8 \\ &= 0.6634 \end{aligned}$$

Therefore,

Probability of correction $\approx 66.3\%$

Assumption: Majority of neighboring pixels are noise-free.

d. Total number of noisy pixels

Image size:

$$1024 \times 1024 = 1,048,576$$

Noise density:

$$0.05$$

Noisy pixels

$$\begin{aligned} 1,048,576 \times 0.05 \\ = 52,428.8 \end{aligned}$$

$\approx 52,429$ noisy pixels

e. Expected noisy pixels after filtering

Assume median filter removes **95%** of noisy pixels.

Remaining

$$\begin{aligned} 52,429 \times 0.05 \\ = 2,621 \end{aligned}$$

Approximately 2,621 noisy pixels remain.

f. If Gaussian noise is present

Instead of a median filter, use:

- Gaussian Filter
- Wiener Filter

because Gaussian noise affects all pixels slightly rather than replacing them with extreme black or white values.

Question 2: Discretization of a Continuous Image

a. Conversion into digital image

Two steps are used:

1. **Sampling** – Convert continuous coordinates into pixels.

2. **Quantization** – Convert continuous intensity into discrete intensity levels.

b. Spatial variation pattern

The intensity changes as a **sinusoidal wave** along the **x-direction**.

Along the y-direction, intensity remains constant.

c. Minimum sampling requirement

According to the **Nyquist Sampling Theorem**,

Sampling frequency should be

$$f_s \geq 2f_{max}$$

where

f_{max} = highest frequency of the sinusoidal signal.

d. If sampled below Nyquist rate

Aliasing occurs.

Effects:

- Distorted image
 - False patterns
 - Loss of details
-

e. Suitable quantization

Use **8-bit quantization**

Number of gray levels

$$2^8 = 256$$

Intensity range

0–255

f. Effect of quantization

- Better image quality with more bits
 - 8-bit requires moderate storage
 - Higher bit depth improves intensity accuracy
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Question 3: Edge Detection in Thin Line Images

a. Comparison

Feature	Sobel	Prewitt	Canny
Noise resistance	Good	Moderate	Excellent
Accuracy	Good	Good	Best
Thin edge detection	Moderate	Moderate	Excellent
Computational cost	Low	Low	High

Best detector: Canny

b. Sobel gradient response

Standard Sobel masks

Horizontal

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Vertical

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

For a horizontal edge,

$$\begin{aligned} G_x &= 0 \\ G_y &= 1020 \end{aligned}$$

Gradient magnitude

$$\begin{aligned} G &= \sqrt{G_x^2 + G_y^2} \\ &= \sqrt{0 + 1020^2} \\ &= 1020 \end{aligned}$$

c. Gradient magnitude and direction

- Gradient magnitude indicates **edge strength**.
 - Gradient direction indicates **edge orientation**.
-

d. Role of Gaussian smoothing

Gaussian smoothing:

- Removes noise
 - Reduces false edges
 - Improves edge detection before gradient calculation
-

e. Effect on thin lines

Too much Gaussian smoothing can

- Blur thin lines
 - Shift edge position
 - Reduce localization accuracy
-

f. Reducing Canny threshold to half

Effects:

- More edges detected
 - Weak edges become visible
 - False edges also increase
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Question 4: Feature Extraction for Handwritten Digit Recognition

a. Working principles

SIFT

- Detects scale-space keypoints
 - Uses Difference of Gaussian
 - Rotation invariant
-

SURF

- Faster than SIFT
 - Uses Hessian matrix
 - Approximate descriptor
-

ORB

- Uses FAST keypoint detector
- Uses BRIEF descriptor
- Fast and suitable for real-time systems

b. Comparison

Property	SIFT	SURF	ORB
Scale invariant	Yes	Yes	Partial
Rotation invariant	Yes	Yes	Yes
Speed	Slow	Medium	Fast
Real-time	No	Moderate	Yes

c. Image rotated by 45°

Reliable methods:

- SIFT
- SURF
- ORB

All provide rotation invariance.

d. Total keypoints

Average keypoints

120

Images

50

Total

$$\begin{aligned} &120 \times 50 \\ &= 6000 \end{aligned}$$

Answer: 6000 keypoints

e. Suitable dimensionality reduction

Principal Component Analysis (PCA)

Retains about **95%** of useful information while reducing feature dimensions.

f. Benefits

Dimensionality reduction

- Faster training
- Lower memory usage

- Removes redundant features
 - Improves classification efficiency
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Question 5: Morphological Preprocessing

a. Remove white noise dots

Use **Opening**

(Erosion followed by Dilation)

b. Fill gaps inside characters

Use **Closing**

(Dilation followed by Erosion)

c. Difference

Opening

Removes small white noise

Smooths object boundary

Closing

Fills small holes

Connects broken parts

d. Effect of 3×3 structuring element

- Removes small noise
 - Preserves most character shapes
 - May slightly thin narrow strokes
-

e. Risk of large structuring element

- Removes important details
- Thickens characters excessively
- Merges nearby characters
- Reduces recognition accuracy

